

Hybrid Mechanistic, Bayesian Inference & Optimization System for Passion-Fruit Pectin Extraction

Research & Experimental Design Blueprint

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1. Executive Summary

This document synthesizes the complete research architecture for building a digital twin of passion-fruit pectin extraction. The architecture bridges a strict mechanistic kinetic model with a Bayesian Optimal Experimental Design (OED) framework. To solve the fundamental bottleneck of expensive laboratory analytics, the system employs a hierarchical proxy measurement model, allowing dense time-series data collection using cost-effective chemometric and rheological sensors.

The sequence of this research project is explicitly defined as:

Mechanistic Model → Synthetic Recovery → OED → Proxy Calibration
→ Physical Experiment → Bayesian Update → Optimization

2. Generalized Methodological Framework

While this document details the specific application of pectin acid hydrolysis, the underlying architecture represents a highly generalizable Hybrid Mechanistic–Statistical Experimental Learning System. This methodology applies broadly to bioprocessing, materials synthesis, and reaction engineering where experiments are expensive and optimization is critical.

2.1. The Three Pillars of System Intelligence

The framework explicitly separates three distinct computational responsibilities:

Mechanistic Model → What can happen? (Physics/Chemistry)
Bayesian Inference → What do we currently believe? (Uncertainty)
OED / Active Learning → What should we measure next? (Information)

Machine learning surrogates serve strictly to make these three operations computationally affordable.

2.2. The Nine Core Components

The complete conceptual architecture consists of nine integrated components:

1. **Mechanistic Process Model:** Represents the underlying physical system ($x_{\{t+1\}} = f(x_t, u_t, \theta)$), providing interpretability, extrapolation structure, conservation laws, and a place to encode prior scientific knowledge.
2. **Identifiability Analysis:** Determines what can actually be learned from available measurements using sensitivity matrices ($S = \frac{\partial y}{\partial \theta}$).

3. **Optimized Experimental Design (OED):** Selects experiments that maximize information gain ($x^* = \arg \max_x \mathcal{I}(x)$) using formal optimality criteria (e.g., D-optimality).
4. **Hierarchical Proxy Measurement Model:** Models the entire observation chain (True State $\rightarrow$ Preparation $\rightarrow$ Proxy $\rightarrow$ Observation) rather than assuming perfect sensors, turning expensive assays into calibration anchors for dense, cheap measurements.
5. **Bayesian Parameter Inference:** Infers the full posterior probability ($p(\theta \mid D)$) rather than point estimates, quantifying uncertainty in both parameters and future predictions ($p(y^* \mid x^*, D)$).
6. **Surrogate ML Modeling:** Learns the expensive forward mapping ($\hat{f}(x, \theta) \approx f(x, \theta)$) to accelerate inference and optimization.
7. **Optimization Under Uncertainty:** Replaces standard expected-value maximization with robust decision-making (e.g., maximizing the probability of an acceptable outcome subject to quality/reliability constraints).
8. **Sequential Active Learning:** Closes the loop. The system does not run once; it continuously iterates: Model $\rightarrow$ Optimal Experiment $\rightarrow$ Data $\rightarrow$ Bayesian Update $\rightarrow$ Next Experiment.
9. **Model Validation & Discrepancy:** Acknowledges that $y = f(x, \theta) + \delta(x) + \varepsilon$, formally modeling the systematic structural mismatch ($\delta(x)$) between the mathematical equations and physical reality to enable true model criticism.

2.3. The Core Abstraction

The deepest abstraction of this system is not “use ML to optimize a reactor.” It is:

Build a mechanistic model, quantify what you don’t know, design the cheapest experiments that maximally reduce that uncertainty, use inexpensive measurements wherever possible through calibrated observation models, and close the loop by updating the model and selecting the next experiment.

3. Mechanistic Reactor Model (V1.3)

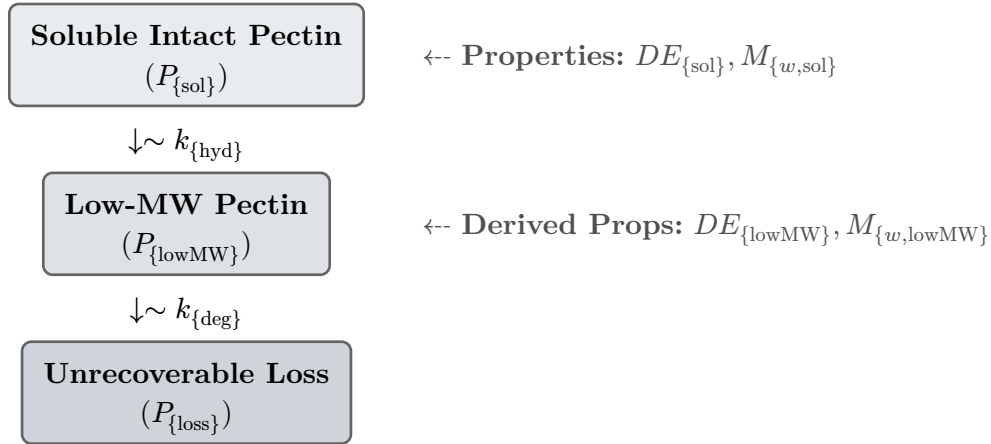
3.1. Compartmental Cascade

The extraction is modeled as a 4-pool cascading mass transfer system. The cascade structure forces a critical structural tradeoff between extraction speed and product degradation.

Matrix Pectin

$(P_{\{\text{matrix}\}})$

$\downarrow \sim k_{\{\text{ext}\}}$



3.2. Kinetic Equations (Mass & Moments)

The state equations combine strict normalized mass conservation [$\text{kg}/\text{kg}_{\{\text{dry,feed}\}}$] with mass-weighted moments to track intensive properties (M_w, DE) dynamically.

$$\frac{dP_{\{\text{sol}\}}}{dt} = k_{\{\text{ext}\}} P_{\{\text{matrix}\}} - k_{\{\text{hyd}\}} P_{\{\text{sol}\}}$$

$$\frac{dP_{\{\text{lowMW}\}}}{dt} = k_{\{\text{hyd}\}} P_{\{\text{sol}\}} - k_{\{\text{deg}\}} P_{\{\text{lowMW}\}}$$

Kinetics are parameterized via an Arrhenius reference-state formulation ($T_{\{\text{ref}\}} = 353.15 \text{ K}$, $pH_{\{\text{ref}\}} = 2.0$) to dramatically improve the numerical conditioning of the Bayesian inference.

3.3. Parameter Ontology

To avoid mathematical ambiguity, all parameters are strictly assigned an ontology classification.

Symbol	Meaning	Units	V1 Range	Classification
T	Extraction Temperature	$^{\circ}\text{C}$	45 – 100 $^{\circ}\text{C}$	Controlled Process Variable
pH	Kinetic Acidity	pH	1.2 – 4.5	Controlled Process Variable
t	Extraction Time	min	5 – 120 min	Controlled Process Variable
$d_{\{50\}}$	Particle Size	μm	149 – 600 μm	Measured State
$M_{\{w,\text{matrix}\}}$	MW of matrix pectin	Da	6.54×10^5	Informative Prior
$k_{\{i,\text{ref}\}}$	Reference Rates	$\text{min}^{\{-1\}}$	TBD	Calibrated

$E_{\{i\}}$	Activation Energies	J/mol	TBD	Calibrated
$n_{\{i\}}$	Acidity Orders	-	1.0	Fixed V1
α	Particle Size Scaling	-	0.5 – 2.0	Calibrated

4. Structural Identifiability & OED

The serial cascade induces a severe identifiability problem: a fast extraction/fast degradation trajectory looks mathematically identical to a slow extraction/slow degradation trajectory if observed only at the process endpoint.

4.1. Bayesian Optimal Experimental Design (OED)

By computing the Fisher Information Matrix (FIM) over the 9-dimensional parameter space, we designed an exact 10-run protocol that minimizes the determinant of the posterior covariance matrix (D-optimal design).

The OED analysis strongly favors time-resolved observation, particularly at early and intermediate residence times, to break the $k_{\{\text{hyd}\}} \leftrightarrow k_{\{\text{deg}\}}$ correlation. It pushes the boundary conditions of the operating space (e.g., 95 °C/pH 1.5 vs 50 °C/pH 3.0). For the final 10-run protocol utilizing 4-point time-resolved tracking, the selected protocol substantially improves the conditioning of the inverse problem, eliminating near-null sensitivity directions and achieving a condition number of $\kappa \approx 27.0$.

The explicit 10-run protocol defined by the solver is as follows (distribute across 2 baseline feedstock batches):

Run	T (°C)	pH	$d_{\{50\}}$ (μm)	Sample Times (min)
1	95	1.5	600	15, 45, 90, 120
2	50	1.5	150	15, 45, 90, 120
3	95	2.5	150	15, 45, 90, 120
4	50	1.5	600	15, 45, 90, 120
5	95	2.0	600	15, 45, 90, 120
6	50	2.0	150	15, 45, 90, 120
7	95	2.5	600	15, 45, 90, 120
8	50	1.5	300	15, 45, 90, 120
9	95	3.0	150	15, 45, 90, 120
10	95	2.0	300	15, 45, 90, 120

5. Hierarchical Measurement Architecture

The time-resolved OED campaign requires 6 samples per run. Performing conventional SEC-MALLS (for M_w), Titration (for DE), and mHDP Colorimetry (for GalA purity) on every aliquot is economically unviable.

We solve this using a Bayesian Proxy Measurement Model. The physical experiment is segmented into a preparation layer and an observation layer.

5.1. The Preparation Model & Precipitation Bias

Direct measurement of raw reactor liquor is strongly confounded by the extraction matrix (unreacted acid, trace metal ions, simple sugars) and is therefore not the preferred basis for routine proxy measurements. All proxy samples undergo Standardized Alcohol Precipitation.

This is not a neutral step. Ethanol fractionation can selectively recover high- M_w pectin while leaving low- M_w fragments in solution. We model this recovery bias as $\eta_{\{\text{precipitation}\}} = f(M_w, DE)$. The proxy must be calibrated on the exact same precipitated material to absorb this physical bias natively.

5.2. The Proxy Observation Model

We deploy two ultra-fast, low-cost sensors on the standardized precipitate:

1. **FTIR/DRIFTS Spectroscopy:** A chemometric PLS model maps the full mid-IR spectrum to DE and GalA fraction.
2. **Capillary Viscometry:** The flow time ($t_{\{\text{flow}\}}$) at a strictly recorded concentration ($C_{\{\text{pellet}\}}$) provides a low-cost MW proxy calibrated directly against HPSEC-MALLS on the same passion-fruit pectin material, avoiding reliance on universal Mark-Houwink constants.

5.3. Proxy Calibration & Acceptance Criteria

To ensure the proxy models generalize, the calibration samples must span the entire operating space. We execute 5 dedicated reactor runs, taking 6 time-resolved aliquots per run (Total $N = 30$).

Run	T (°C)	pH	$d_{\{50\}}$ (μm)	Targeted Domain	Expected State
C1	50	3.0	600	Absolute Min Severity	High M_w , High DE
C2	95	1.5	150	Absolute Max (Stress)	Low M_w , Low DE
C3	75	2.2	300	Interior Trajectory	Moderate M_w , Mod DE
C4	95	3.0	300	High T / Low Acid	Moderate M_w , High DE

C5	50	1.5	150	Low T / High Acid	High M_w , Low DE
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Sampling Schedule: Withdrawn at $t = 10, 25, 45, 70, 95, 120$ minutes.

Because time points within a single run are autocorrelated, standard cross-validation will overfit. We mandate Leave-One-Run-Out (LORO) Cross-Validation. **Acceptance Metrics:**

1. **RMSEP:** Must be $< 5\%$ absolute for DE/GalA , and < 50 kDa for M_w .
2. **Stratified RMSEP:** The proxy must not fail catastrophically at extremes (e.g., reported separately for low/med/high ranges).
3. **MAE & Bias:** Residuals must be centered at zero.
4. **Uncertainty Coverage:** The proxy's 95% CI must contain the true reference value 95% of the time on unseen runs ($P(y_{\{\text{true}\}} \in CI_{\{95\%, \text{proxy}\}})$).

5.4. Bayesian Integration

The measurement hierarchy resolves into a clean Directed Acyclic Graph (DAG):

$$\text{Kinetic ODE} \rightarrow y_{\{\text{true}\}} \rightarrow y_{\{\text{prepared}\}} \rightarrow z_{\{\text{observed}\}}$$

Where $y = (Y, DE, M_w, X_{\{\text{GalA}\}})$ and $z = (\text{FTIR}, t_{\{\text{flow}\}}, C_{\{\text{ppt}\}}, C_{\{\text{pellet}\}})$. Then the posterior over kinetic parameters is obtained by marginalizing the latent physicochemical state:

$$p(\theta \text{ mid } z, x) \propto p(\theta) \int p(z \text{ mid } y)p(y \text{ mid } \theta, x)dy$$

6. Physical Implementation & Minimum Viable Laboratory (MVP)

The hierarchical measurement architecture directly enables a decentralized, low-capital laboratory strategy. By mathematically separating high-frequency proxies from low-frequency ground-truth anchors, the physical execution requires minimal in-house capital expenditure.

6.1. Equipment Sourcing Strategy

The project relies on a hybrid execution model, partnering with university core facilities for spectral and chromatographic capabilities:

1. **In-House (Own / Operate):**
 - 1.0 – 2.0 L jacketed glass reactor (sized to prevent S:L ratio drift during 120 mL cumulative sampling).
 - Continuous RTD temperature logging and pH meters.

- Standardized precipitation equipment (centrifuge, vacuum oven).
 - Calibrated analytical balance (Critical: drives the entire mass-basis of the kinetic ODE).
 - Capillary viscometer and UV-Vis spectrophotometer.
2. **University / Core Facility Access:**
- ATR-FTIR Spectrometer (High-frequency chemical proxy).
 - HPSEC-MALLS (Low-frequency M_w anchor).

Expensive automated bioreactors, inline FTIR sensors, and sophisticated rheometers are explicitly deferred.

6.2. Analytical Discipline & Traceability

The Bayesian inference engine requires an unbroken relational chain linking the final measurement back to the thermodynamic history of the extraction.

The primary key for all data ingestion is the composite **SampleID** (e.g., B002-R07-S03). This uniquely binds a physical pellet to:

$$t_{\{\text{aliquot}\}} + T(t) + pH(t) + m_{\{\text{liquor}\}} + m_{\{\text{pellet}\}}$$

If this chain is broken, the sample may no longer be suitable for kinetic inference because its process history cannot be assigned reliably.

7. Conclusion & Execution Map

The mechanistic model has been mathematically verified, structural identifiability resolved via D-optimal design, and the measurement architecture strictly defined. By explicitly modeling sample preparation and measurement uncertainty as part of the Bayesian observation hierarchy, the project is insulated from the capital costs and physical biases that typically derail bioreactor optimization.

The reactor is specified here not merely as a machine producing pectin, but as an experimental measurement instrument built specifically for identifying a dynamic model.

The immediate physical step is the Phase 1 Proxy Calibration, generating 30 paired observations across 5 OED extreme trajectories to construct the $p(y \mid z)$ measurement likelihoods. Once the chemometric and rheological proxies are validated against the SEC/Titration anchors, the system proceeds to the 10-run OED-selected calibration campaign plus 4 independent validation runs for final parameter closure.